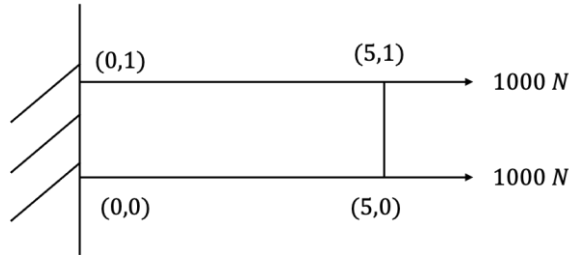


Instructions:

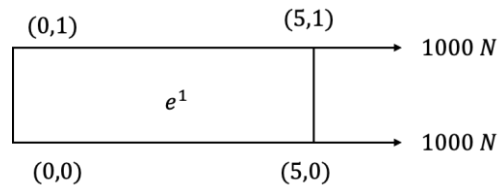
- Submit your assignment as a single PDF file through Canvas.
- Show all your steps and solution procedures.
- Make sure that your solution is well organised and that you are using appropriate headers for each question and sub-question.
- Scanned photos of your handwritten solution are acceptable as long as they are **legible**
- If a question requires plotting, please plot the expressions using any software/language (MATLAB, Excel, Mathematica, etc.) of your choice. However, **hand drawings** will not be accepted.
- You can use any software or calculator to perform any calculations. Using symbolic packages in MATLAB, Mathematica is acceptable as long as the solution steps are well laid out in your submission. **In case you use those packages please attach the script in your submission file as plain text and not a screenshot.**
- While you can discuss the problems with classmates, all submitted work must be your own to conform to academic integrity guidelines.
- Late assignments will not be accepted.

Question 1 (50 Points)

Consider the following structure. Solve for the nodal displacements assuming a plane stress condition (with $t = 0.2$ m) and that the elastic modulus and Poisson's ratio are 100 GPa and 0.3 respectively. Assume that the structure has no body forces.



Solve for the **nodal displacements** using **one linear rectangular element** as shown below:



NB: The coordinates are expressed in m.

Question 2 (20 Points)

Evaluate the Jacobian matrix for a quadrilateral element bounded by (0,0), (2,2), (2,1) and (-1,0.5)

Question 3 (30 Points)

Solve the deep beam tutorial ([Problem 1: Deep Beam](#)) provided from Dr. Adeeb's course textbook on ABAQUS. To show the completion of this exercise show a screenshot of S_{11}